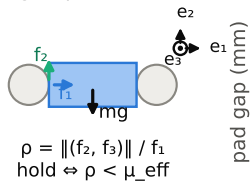


(c) grasp frame & utilization



(d) thin grasp manifold ( $\phi = 0, \theta = 0$ )

